

VIT®

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

ASSESSMENT-01

NAME	:	**GILLEDI SANTHOSH**
REGISTER NUMBER	:	**23MID0233**
COURSE NAME	:	**ADVANCED PYTHON PROGRAMMING**
COURSE CODE	:	**CSI3007**
FACULTY	:	**SHARMILA BANU K**
LAB SLOT	:	**L31+L32+L49+L50**

TASK -1 :

Comparing Ansi Encoding Vs Unicode Encoding.

Aim:

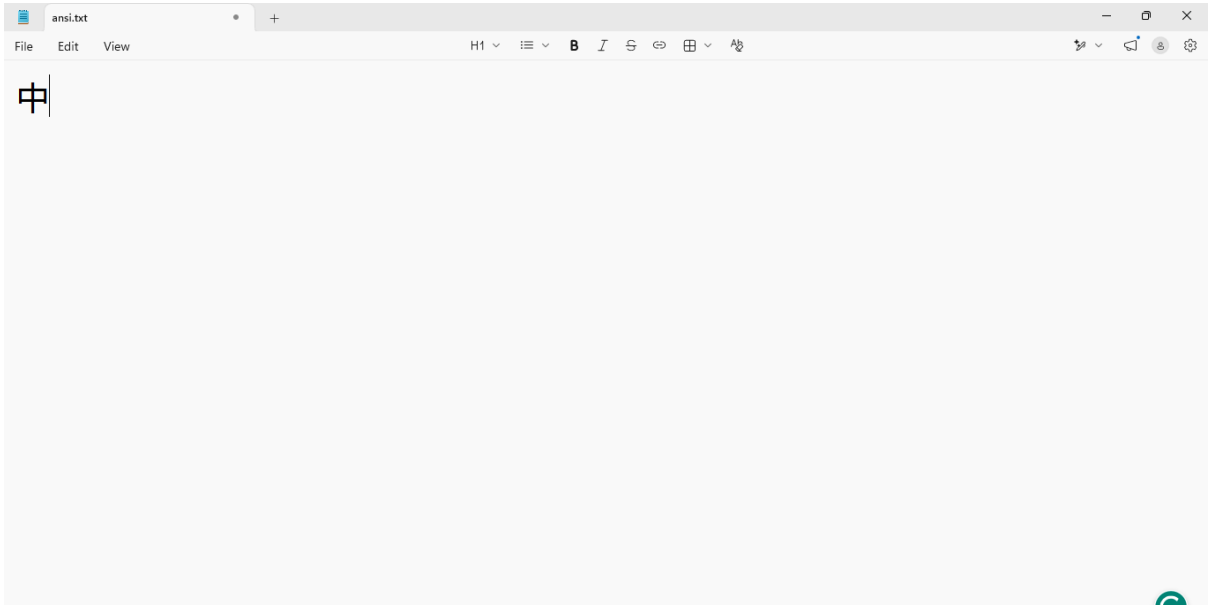
To understand the difference between ANSI encoding and Unicode encoding by saving a Chinese character in different encoding formats using Notepad and observing the results.

1.ANSI Encoding :

ANSI encoding is a traditional 8-bit character encoding system that can represent up to 256 characters in a single code page. Since each code page supports only specific languages, characters not included in the selected code page are replaced with a placeholder symbol such as ?.

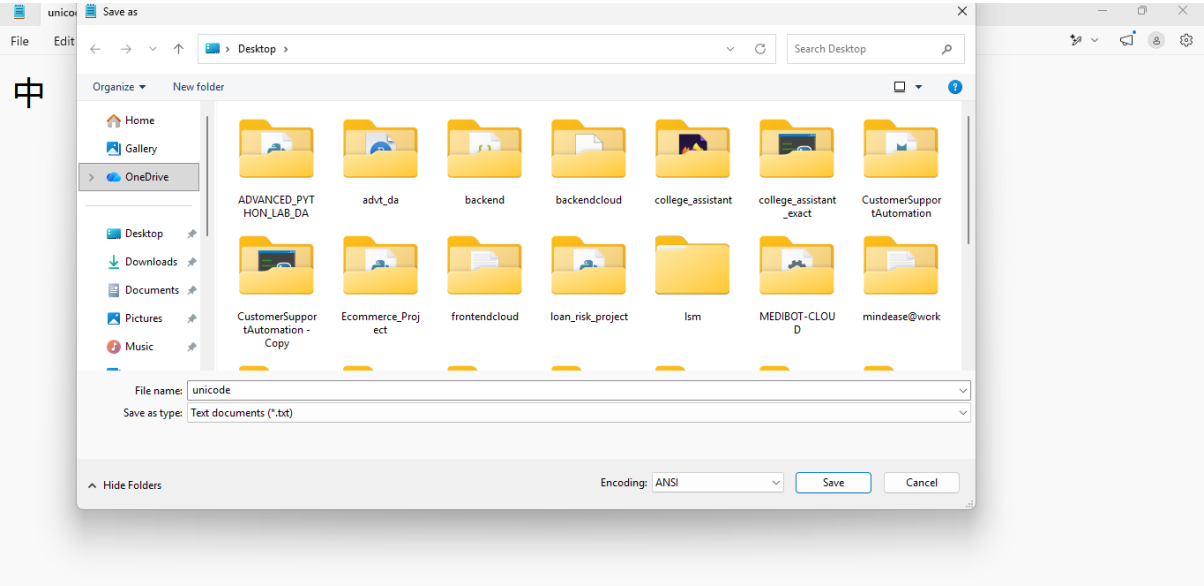
STEPS :

- **Enter a chinese character in the notepad :**



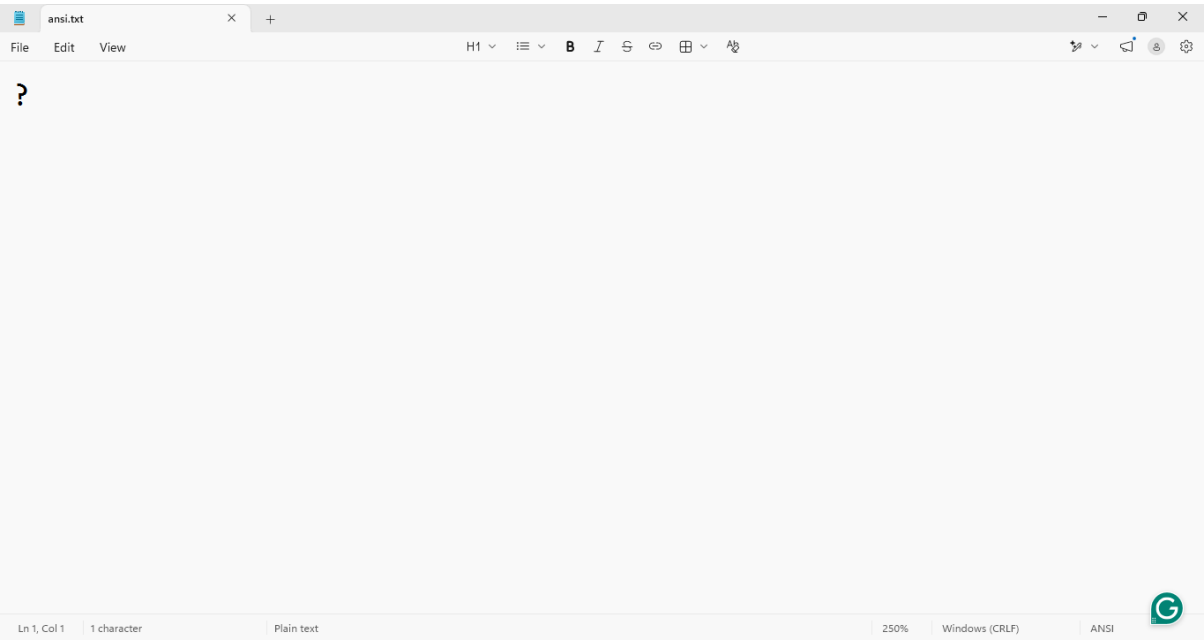
- **Save the File Using ANSI Encoding.**

1. Click File → Save As.
2. Enter a file name (eg : ansi.txt).
3. In the Encoding drop-down list, select ANSI.
4. Click Save.



3.Reopen the File

Close the file and open **ansi.txt** again.The original Chinese character is replaced by a **?**



Observation : The Chinese character "中" is not available in most English or Indian ANSI code pages. Therefore, when the file is saved using ANSI encoding, Notepad cannot assign a valid character code and replaces it with a question mark (?). However, when the file is saved using Unicode (UTF-8), the character is preserved because Unicode assigns it the unique code point U+4E2D.

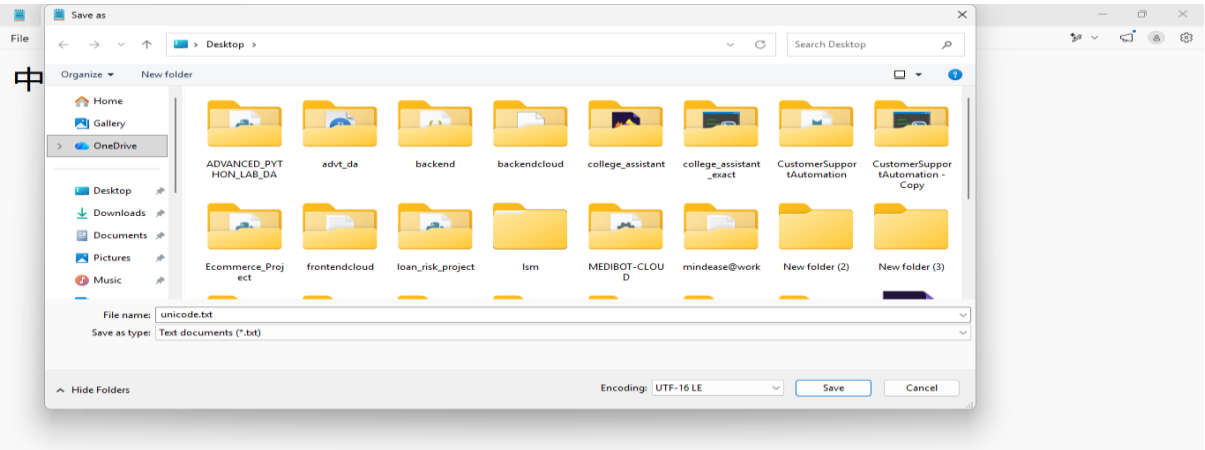
2. Unicode Encoding

Unicode is a universal character encoding standard that assigns a unique code point to every character from almost all languages in the world. It supports over 149,000 characters**,** including letters, numbers, symbols, emojis, and multilingual text. Unicode is commonly implemented using encoding formats such as UTF-8**,** UTF-16**,** and UTF-32**.**

STEPS :

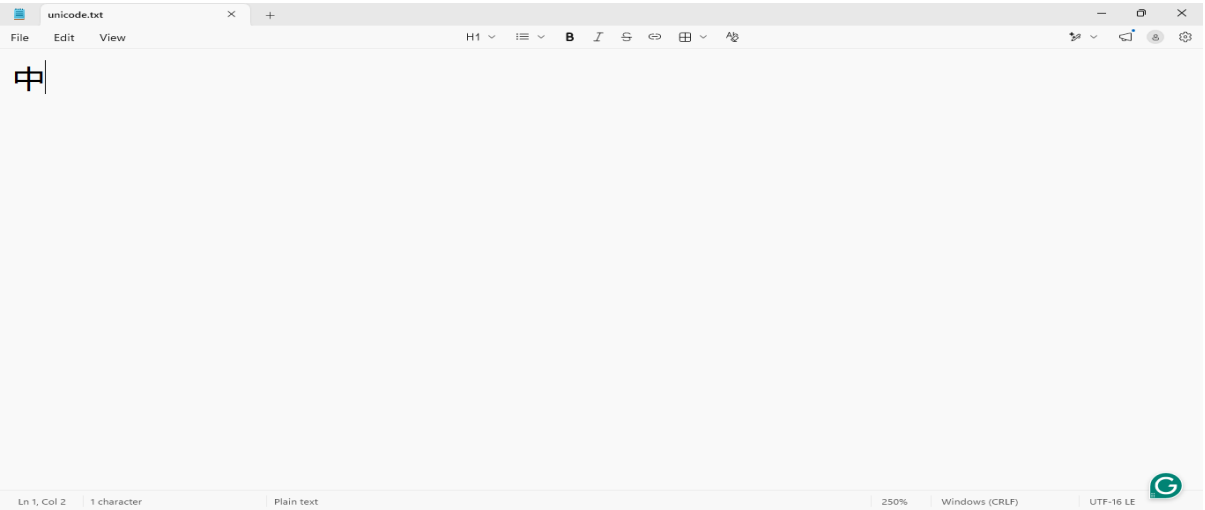
- **Type the Chinese character into Notepad:**

1. Click File → Save As**.**
2. Enter a file name (for example, unicode.txt**).**
3. In the Encoding drop-down list, select UTF-8 **or** UTF-16 LE
4. Click Save**.**



- **2.Reopen the File :**

Close the file and open **unicode.txt** again.The Chinese character 中 is displayed correctly without any changes.



Observation : Unicode assigns a unique code point to every character. Since the Chinese character "中" has the Unicode code point **U+4E2D**, it can be encoded and stored correctly using UTF-8 or UTF-16. As a result, when the file is reopened, the original character is displayed without any changes.

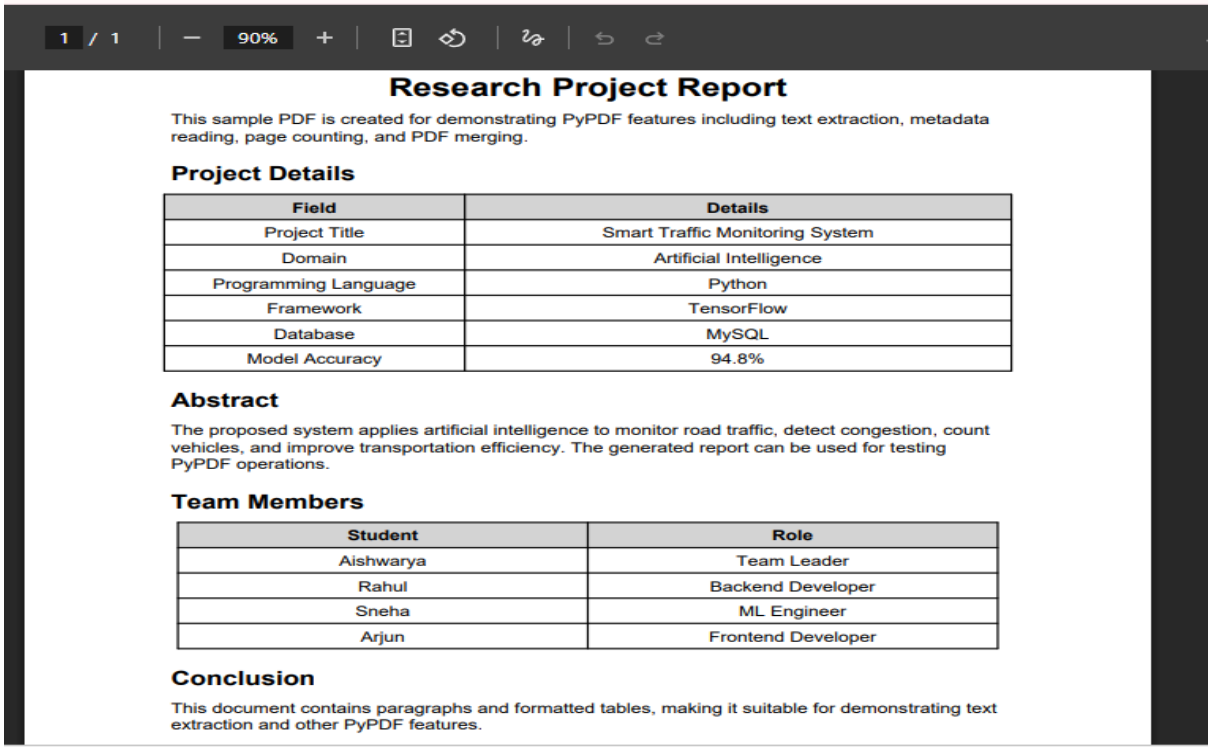
Conclusion

From this experiment, it can be concluded that ANSI and Unicode use different methods to store text. ANSI supports only a limited set of characters, so unsupported characters such as the Chinese character "中" are replaced with a question mark (?) when saved. In contrast, Unicode supports characters from almost all languages by assigning a unique code point to each character. Therefore, Unicode stores and displays multilingual text correctly, making it the preferred encoding standard for modern computers and applications.

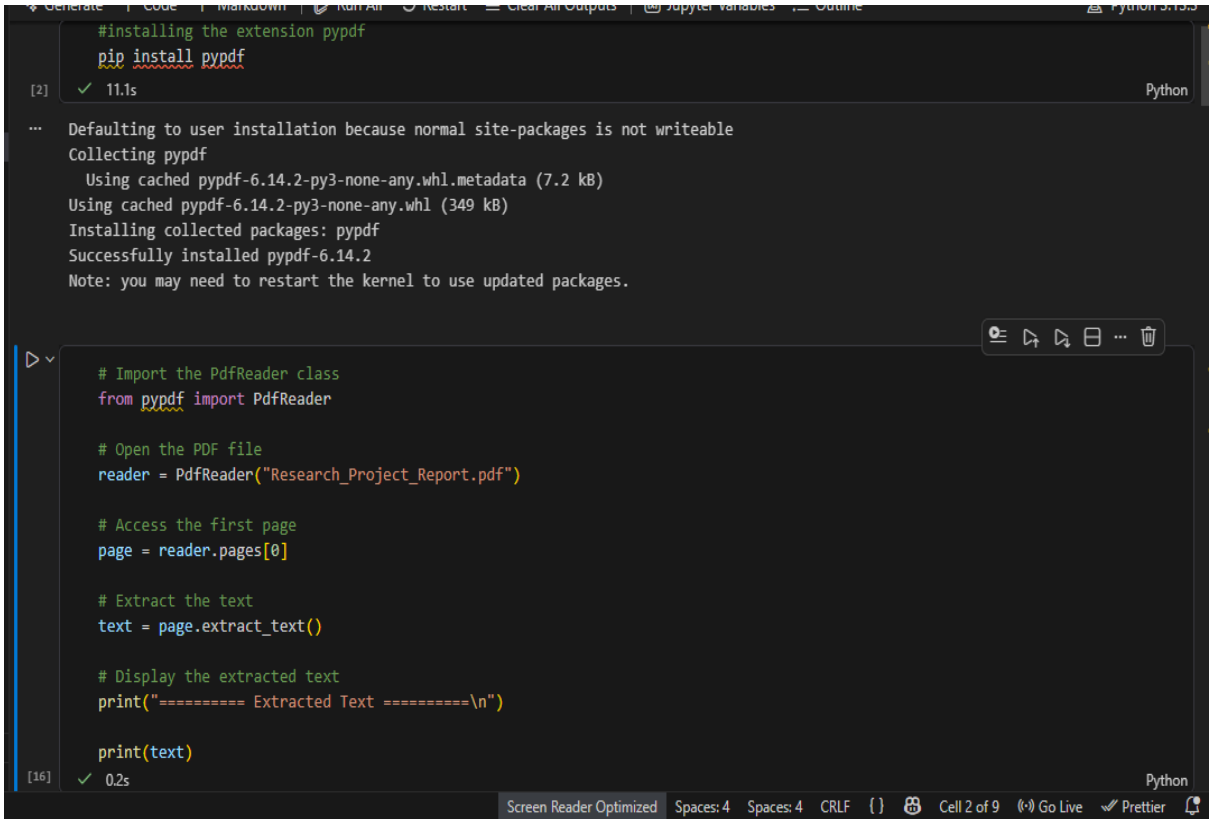
TASK -2 : Exploring python extension for file formats.

1.PyPDF

- **Create a PDF for performing the various operations for exploring the features of PyPDF.**



Feature -1 : Extract Text from a PDF



```
#installing the extension pypdf
pip install pypdf
```

[2] ✓ 11.1s Python

```
Defaulting to user installation because normal site-packages is not writeable
Collecting pypdf
  Using cached pypdf-6.14.2-py3-none-any.whl.metadata (7.2 kB)
Using cached pypdf-6.14.2-py3-none-any.whl (349 kB)
Installing collected packages: pypdf
Successfully installed pypdf-6.14.2
Note: you may need to restart the kernel to use updated packages.
```

```
# Import the PdfReader class
from pypdf import PdfReader

# Open the PDF file
reader = PdfReader("Research_Project_Report.pdf")

# Access the first page
page = reader.pages[0]

# Extract the text
text = page.extract_text()

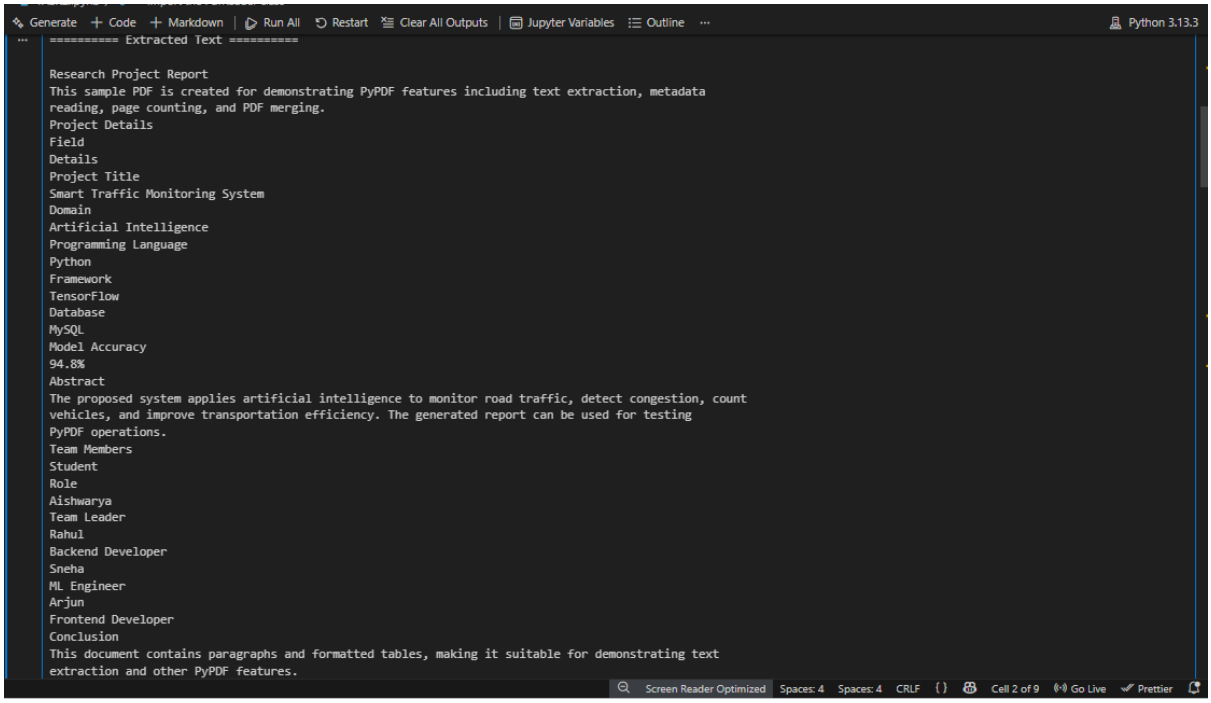
# Display the extracted text
print("========== Extracted Text ==========\n")

print(text)
```

[16] ✓ 0.2s Python

Screen Reader Optimized Spaces: 4 Spaces: 4 CRLF { } Cell 2 of 9 Go Live Prettier

OUTPUT :



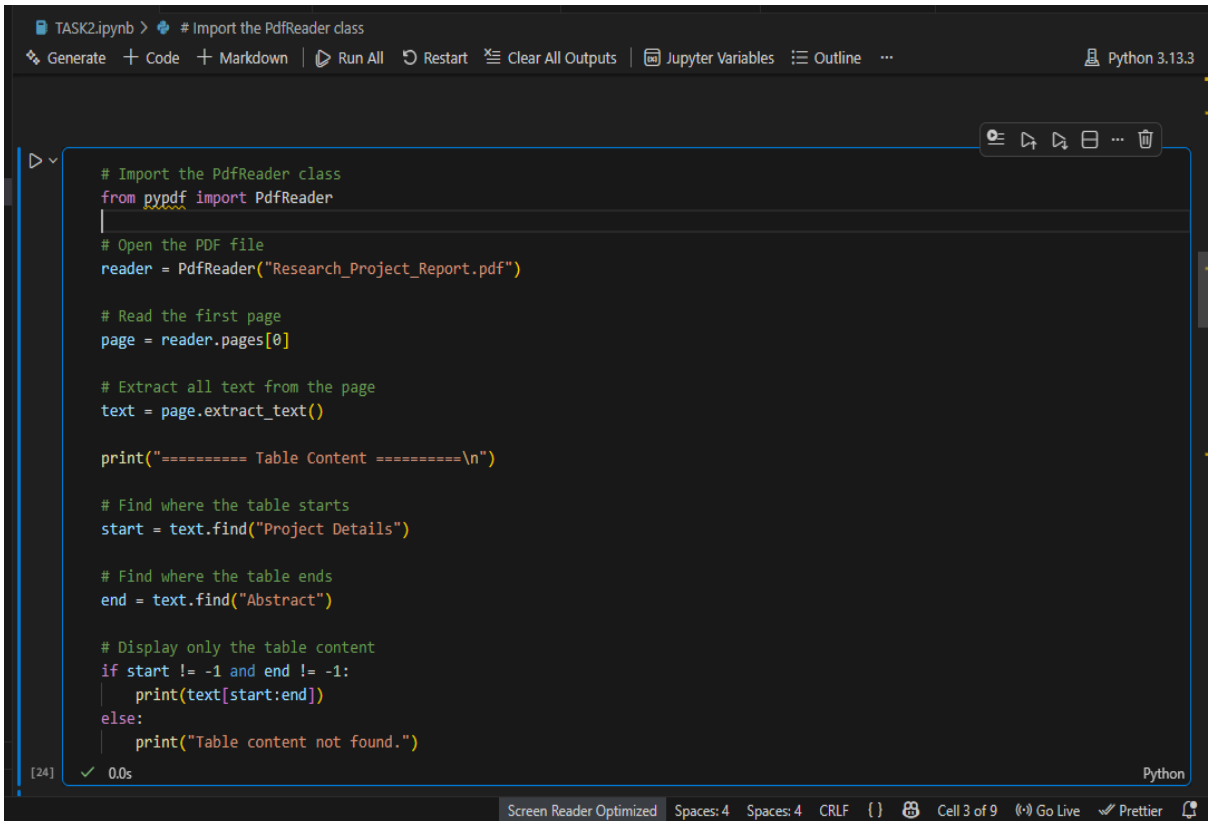
Generate + Code + Markdown | Run All Restart Clear All Outputs | Jupyter Variables Outline ... Python 3.13.3

```
========== Extracted Text ==========

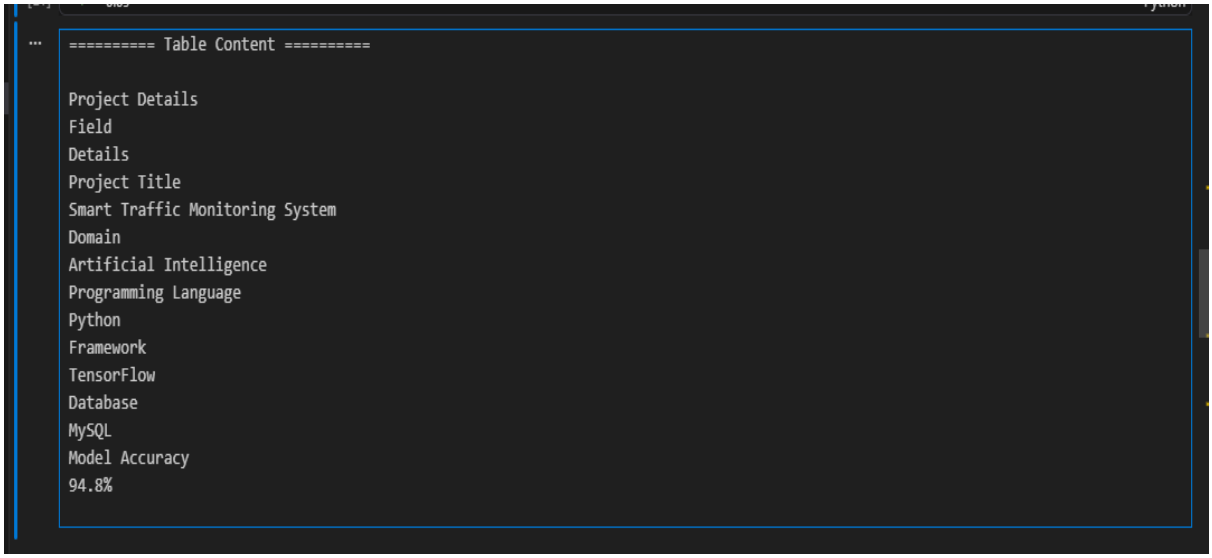
Research Project Report
This sample PDF is created for demonstrating PyPDF features including text extraction, metadata
reading, page counting, and PDF merging.
Project Details
Field
Details
Project Title
Smart Traffic Monitoring System
Domain
Artificial Intelligence
Programming Language
Python
Framework
TensorFlow
Database
MySQL
Model Accuracy
94.8%
Abstract
The proposed system applies artificial intelligence to monitor road traffic, detect congestion, count
vehicles, and improve transportation efficiency. The generated report can be used for testing
PyPDF operations.
Team Members
Student
Role
Aishwarya
Team Leader
Rahul
Backend Developer
Sneha
ML Engineer
Arjun
Frontend Developer
Conclusion
This document contains paragraphs and formatted tables, making it suitable for demonstrating text
extraction and other PyPDF features.
```

Screen Reader Optimized Spaces: 4 Spaces: 4 CRLF { } Cell 2 of 9 Go Live Prettier

Feature -2 : Retrieve Table Content

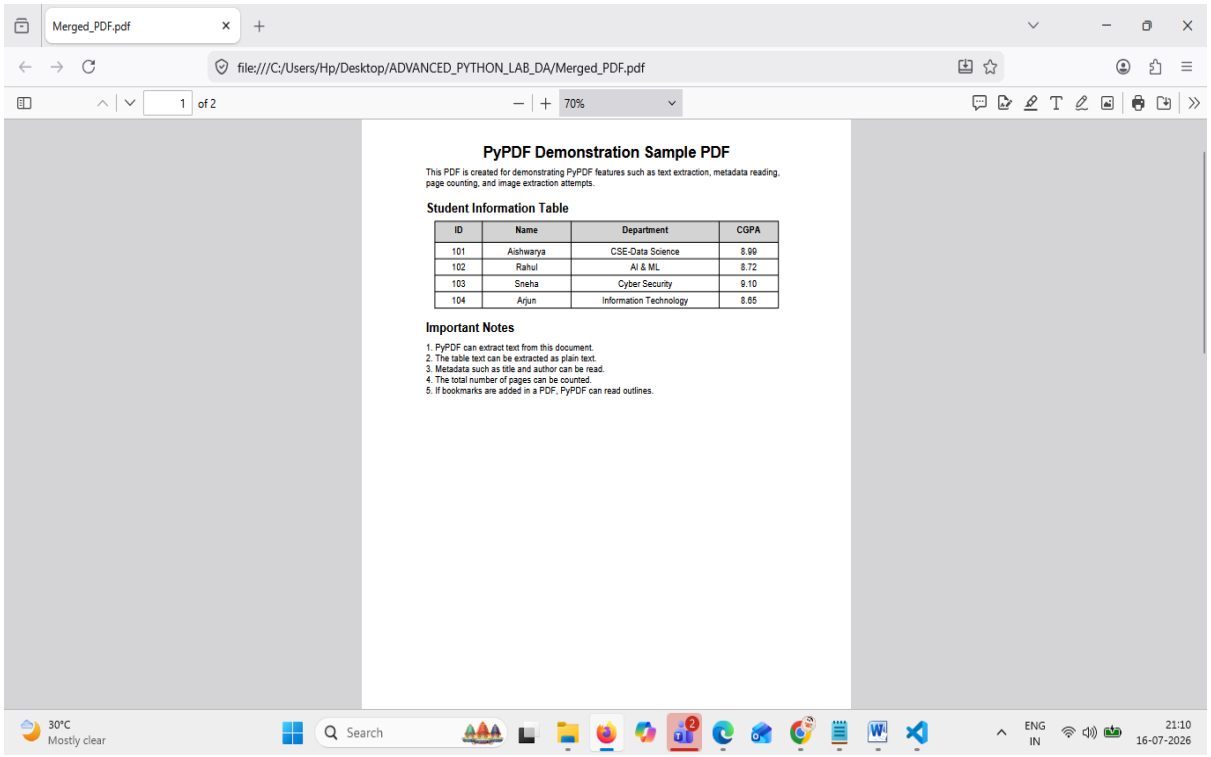


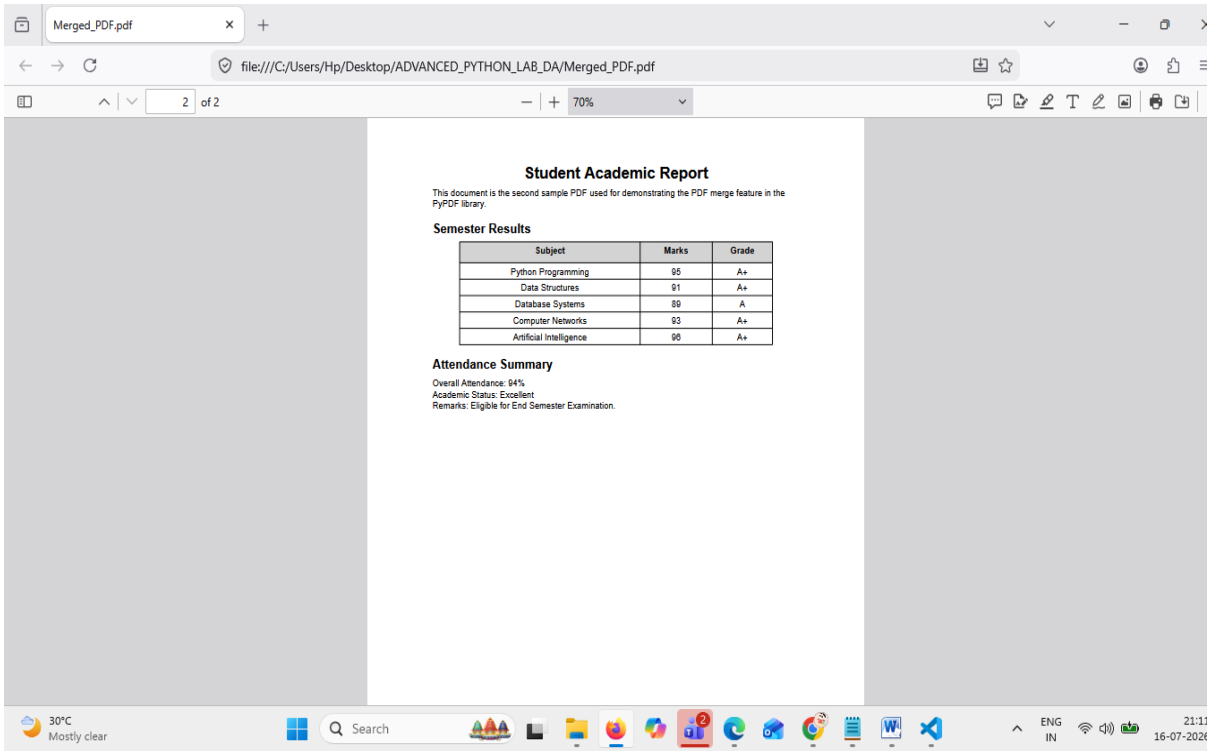
OUTPUT :

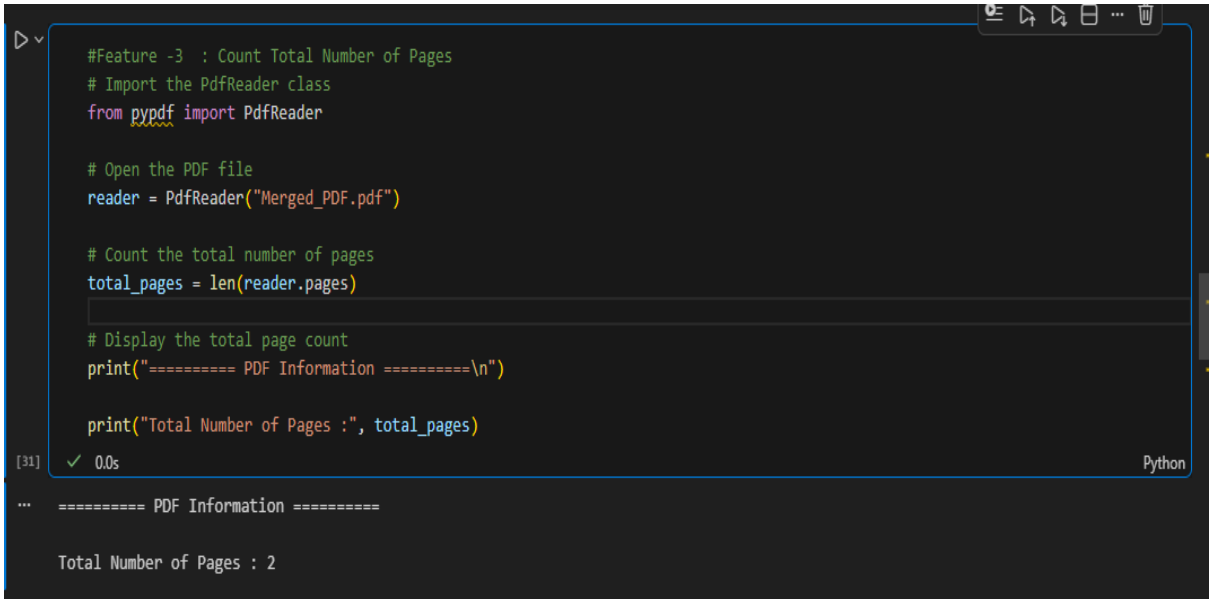


Feature 3 : Count Total Number of Pages

- PDF UPLOADED IS ATTATCHED BELOW:







```python
#Feature -3  : Count Total Number of Pages
# Import the PdfReader class
from pypdf import PdfReader

# Open the PDF file
reader = PdfReader("Merged_PDF.pdf")

# Count the total number of pages
total_pages = len(reader.pages)

# Display the total page count
print("========== PDF Information ==========\n")

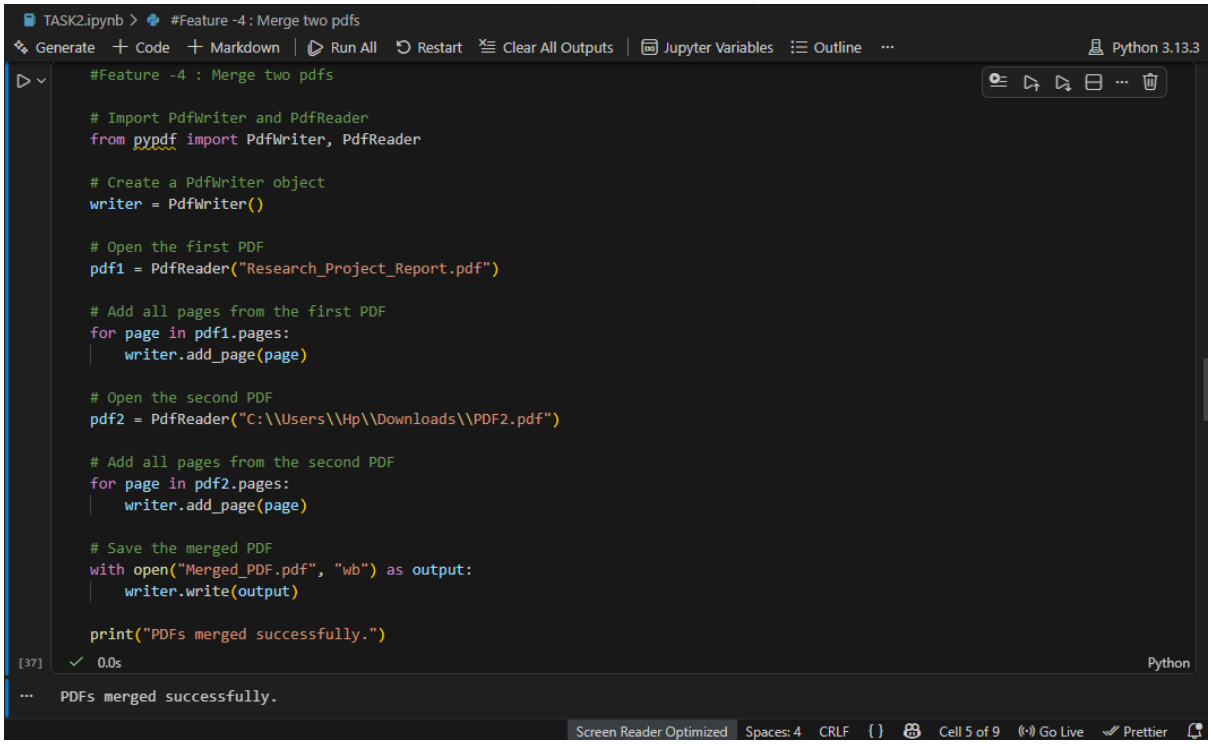
print("Total Number of Pages :", total_pages)
```

[31] ✓ 0.0s Python

```
========== PDF Information ==========

Total Number of Pages : 2
```

Feature -4 : Merging two PDFS.



TASK2.ipynb > #Feature -4 : Merge two pdfs

Generate | Code | Markdown | Run All | Restart | Clear All Outputs | Jupyter Variables | Outline | Python 3.13.3

```python
#Feature -4 : Merge two pdfs

# Import PdfWriter and PdfReader
from pypdf import PdfWriter, PdfReader

# Create a PdfWriter object
writer = PdfWriter()

# Open the first PDF
pdf1 = PdfReader("Research_Project_Report.pdf")

# Add all pages from the first PDF
for page in pdf1.pages:
    writer.add_page(page)

# Open the second PDF
pdf2 = PdfReader("C:\\Users\\Hp\\Downloads\\PDF2.pdf")

# Add all pages from the second PDF
for page in pdf2.pages:
    writer.add_page(page)

# Save the merged PDF
with open("Merged_PDF.pdf", "wb") as output:
    writer.write(output)

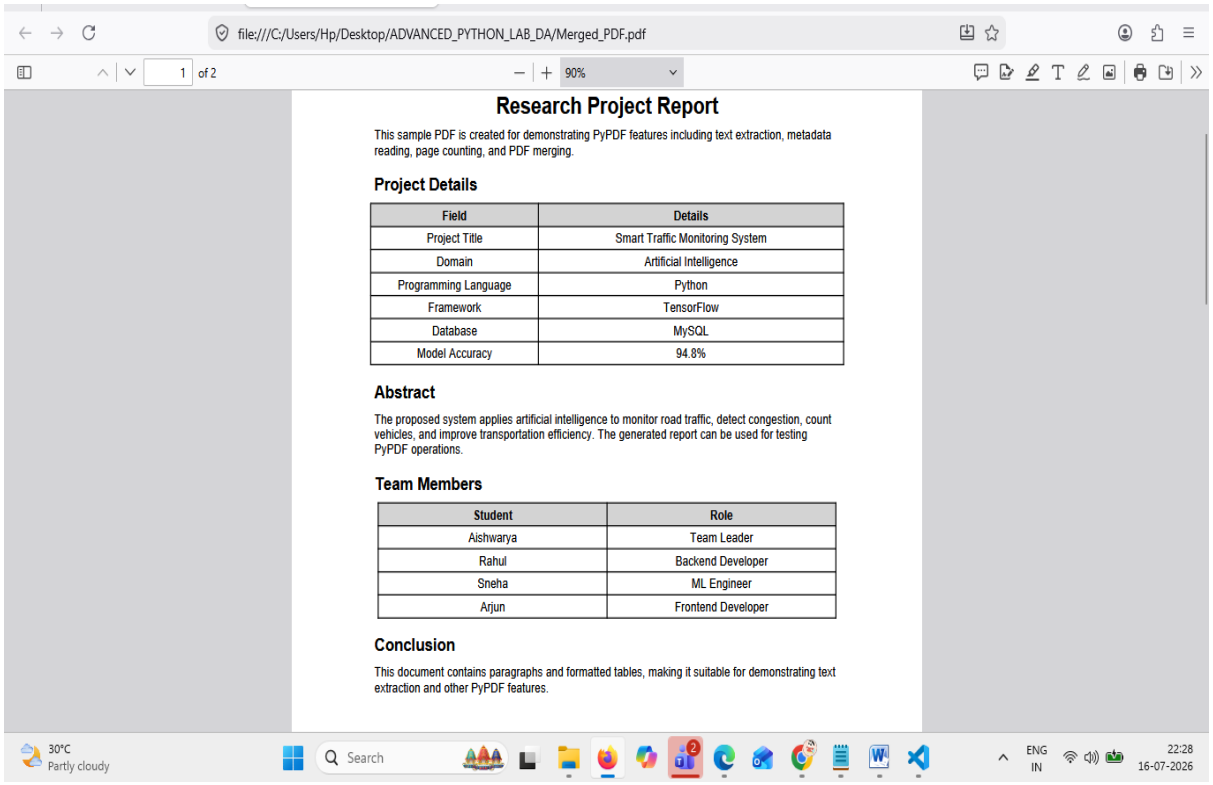
print("PDFs merged successfully.")
```

[37] ✓ 0.0s Python

```
PDFs merged successfully.
```

Screen Reader Optimized | Spaces: 4 | CRLF | { } | Cell 5 of 9 | Go Live | Prettier

OUTPUT :



file:///C:/Users/Hp/Desktop/ADVANCED_PYTHON_LAB_DA/Merged_PDF.pdf

1 of 2 90%

Research Project Report

This sample PDF is created for demonstrating PyPDF features including text extraction, metadata reading, page counting, and PDF merging.

Project Details

Field	Details
Project Title	Smart Traffic Monitoring System
Domain	Artificial Intelligence
Programming Language	Python
Framework	TensorFlow
Database	MySQL
Model Accuracy	94.8%

Abstract

The proposed system applies artificial intelligence to monitor road traffic, detect congestion, count vehicles, and improve transportation efficiency. The generated report can be used for testing PyPDF operations.

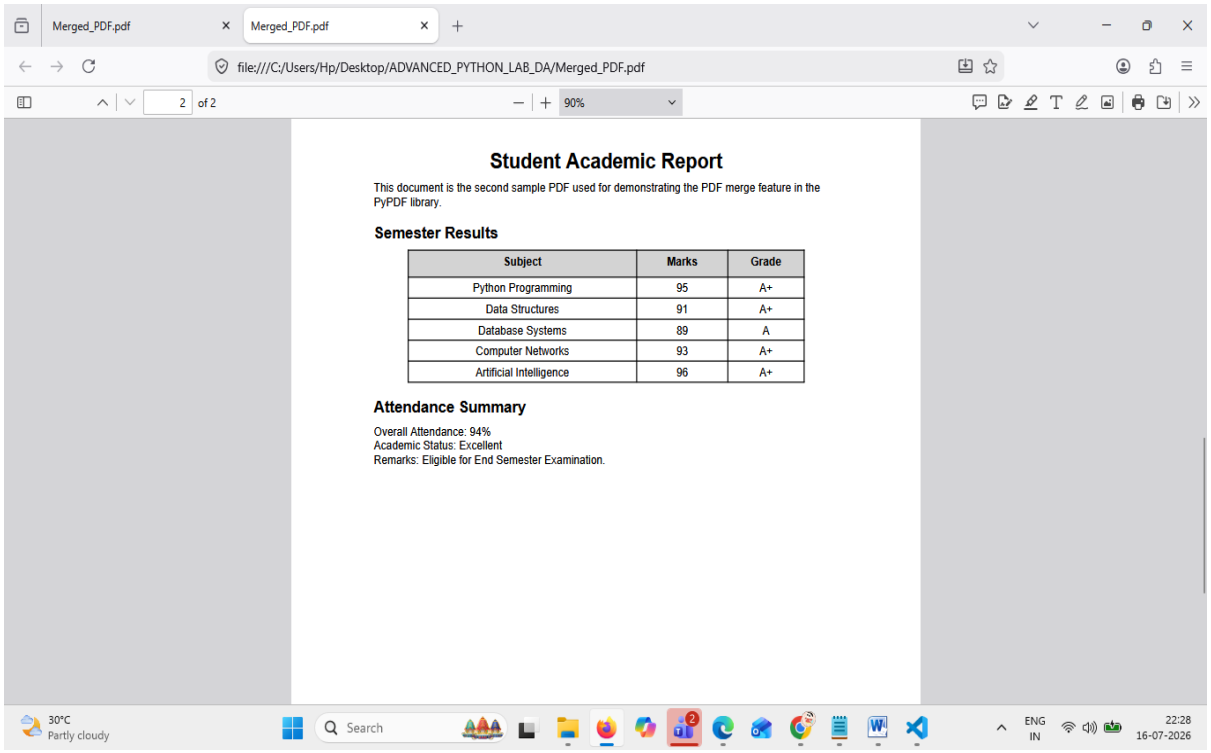
Team Members

Student	Role
Aishwarya	Team Leader
Rahul	Backend Developer
Sneha	ML Engineer
Arjun	Frontend Developer

Conclusion

This document contains paragraphs and formatted tables, making it suitable for demonstrating text extraction and other PyPDF features.

30°C Partly cloudy Search ENG IN 22:28 16-07-2026



Merged_PDF.pdf Merged_PDF.pdf

file:///C:/Users/Hp/Desktop/ADVANCED_PYTHON_LAB_DA/Merged_PDF.pdf

2 of 2 90%

Student Academic Report

This document is the second sample PDF used for demonstrating the PDF merge feature in the PyPDF library.

Semester Results

Subject	Marks	Grade
Python Programming	95	A+
Data Structures	91	A+
Database Systems	89	A
Computer Networks	93	A+
Artificial Intelligence	96	A+

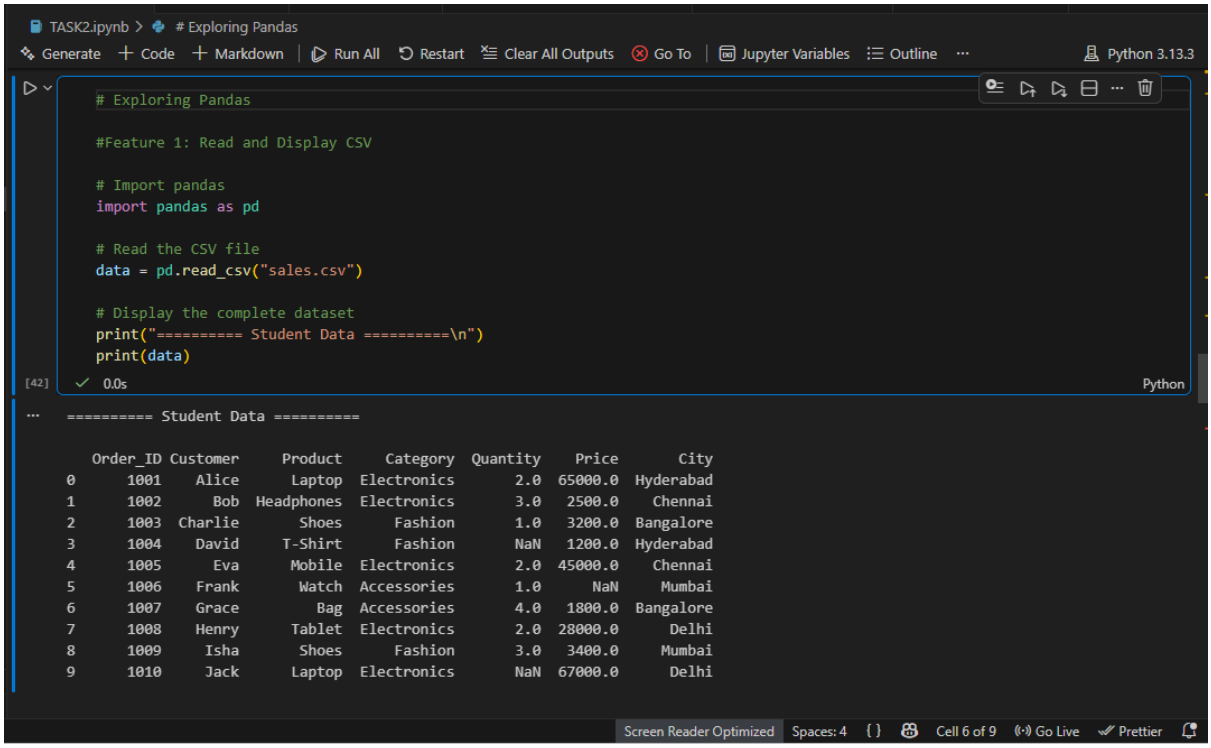
Attendance Summary

Overall Attendance: 94%
Academic Status: Excellent
Remarks: Eligible for End Semester Examination.

30°C Partly cloudy Search ENG IN 22:28 16-07-2026

2.Exploring Pandas for .csv files.

Feature 1: Read and Display CSV



```
# Exploring Pandas

#Feature 1: Read and Display CSV

# Import pandas
import pandas as pd

# Read the CSV file
data = pd.read_csv("sales.csv")

# Display the complete dataset
print("========== Student Data ==========\n")
print(data)
```

```
========== Student Data ==========

   Order_ID Customer     Product     Category  Quantity    Price       City
0      1001    Alice      Laptop  Electronics       2.0  65000.0  Hyderabad
1      1002      Bob  Headphones  Electronics       3.0   2500.0    Chennai
2      1003  Charlie       Shoes      Fashion       1.0   3200.0  Bangalore
3      1004    David     T-Shirt      Fashion       NaN   1200.0  Hyderabad
4      1005      Eva      Mobile  Electronics       2.0  45000.0    Chennai
5      1006    Frank       Watch  Accessories       1.0      NaN     Mumbai
6      1007    Grace         Bag  Accessories       4.0   1800.0  Bangalore
7      1008    Henry      Tablet  Electronics       2.0  28000.0      Delhi
8      1009     Isha       Shoes      Fashion       3.0   3400.0     Mumbai
9      1010     Jack      Laptop  Electronics       NaN  67000.0      Delhi
```

Feature 2: Handling Missing Values



```
#Feature 2: Handle Missing Values

# Import pandas library
import pandas as pd

# Read the CSV file
data = pd.read_csv("sales.csV")

# Display the original dataset
print("========== Original Data ==========\n")
print(data)

# Replace missing values with 0
filled_data = data.fillna(0)

# Display the updated dataset
print("\n========== After Filling Missing Values ==========\n")
print(filled_data)
```

OUTPUT :

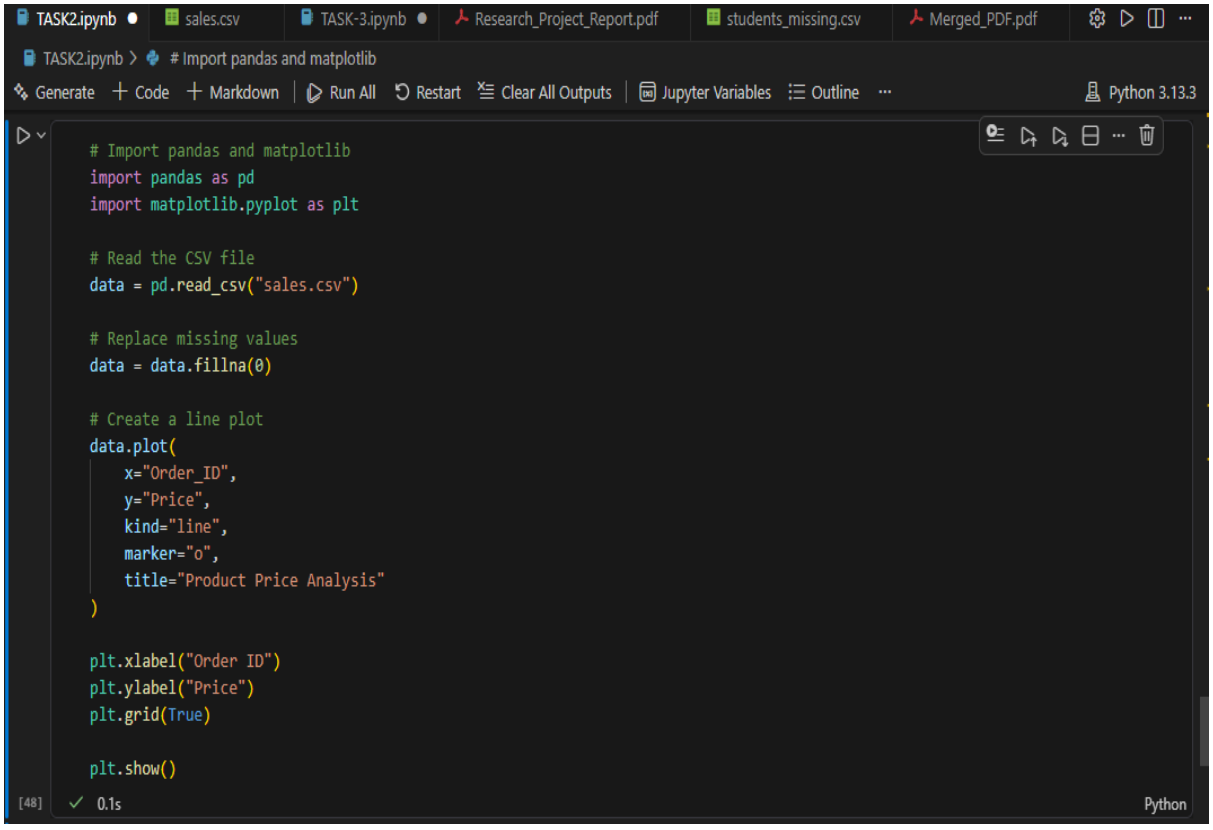
```
========== Original Data ==========

   Order_ID Customer     Product     Category  Quantity    Price       City
0      1001    Alice      Laptop  Electronics       2.0  65000.0  Hyderabad
1      1002      Bob  Headphones  Electronics       3.0   2500.0    Chennai
2      1003  Charlie       Shoes      Fashion       1.0   3200.0  Bangalore
3      1004    David     T-Shirt      Fashion       NaN   1200.0  Hyderabad
4      1005      Eva      Mobile  Electronics       2.0  45000.0    Chennai
5      1006    Frank       Watch  Accessories       1.0      NaN     Mumbai
6      1007    Grace         Bag  Accessories       4.0   1800.0  Bangalore
7      1008    Henry      Tablet  Electronics       2.0  28000.0      Delhi
8      1009     Isha       Shoes      Fashion       3.0   3400.0     Mumbai
9      1010     Jack      Laptop  Electronics       NaN  67000.0      Delhi

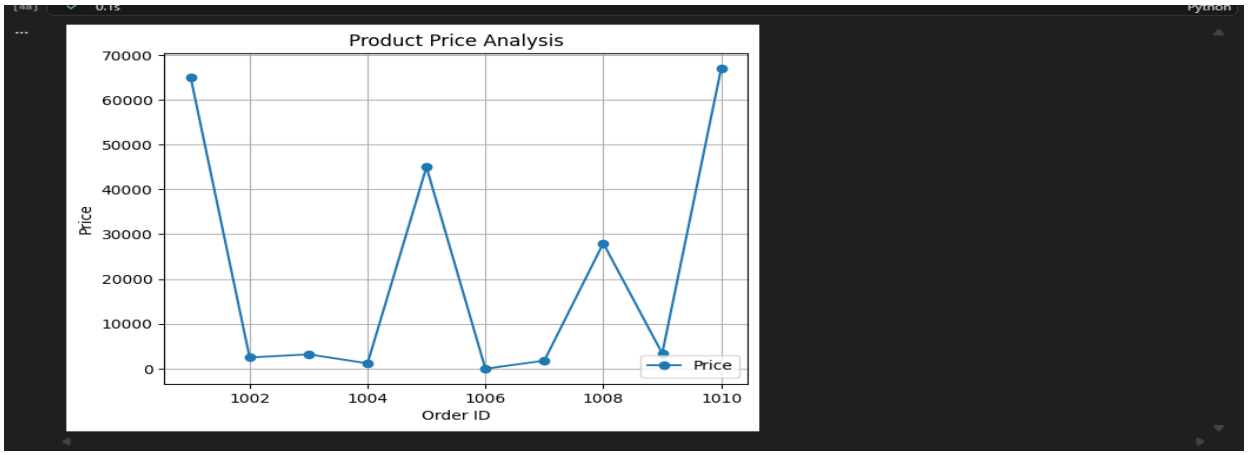
========== After Filling Missing Values ==========

   Order_ID Customer     Product     Category  Quantity    Price       City
0      1001    Alice      Laptop  Electronics       2.0  65000.0  Hyderabad
1      1002      Bob  Headphones  Electronics       3.0   2500.0    Chennai
2      1003  Charlie       Shoes      Fashion       1.0   3200.0  Bangalore
3      1004    David     T-Shirt      Fashion       0.0   1200.0  Hyderabad
4      1005      Eva      Mobile  Electronics       2.0  45000.0    Chennai
5      1006    Frank       Watch  Accessories       1.0      0.0     Mumbai
6      1007    Grace         Bag  Accessories       4.0   1800.0  Bangalore
7      1008    Henry      Tablet  Electronics       2.0  28000.0      Delhi
8      1009     Isha       Shoes      Fashion       3.0   3400.0     Mumbai
9      1010     Jack      Laptop  Electronics       0.0  67000.0      Delhi
```

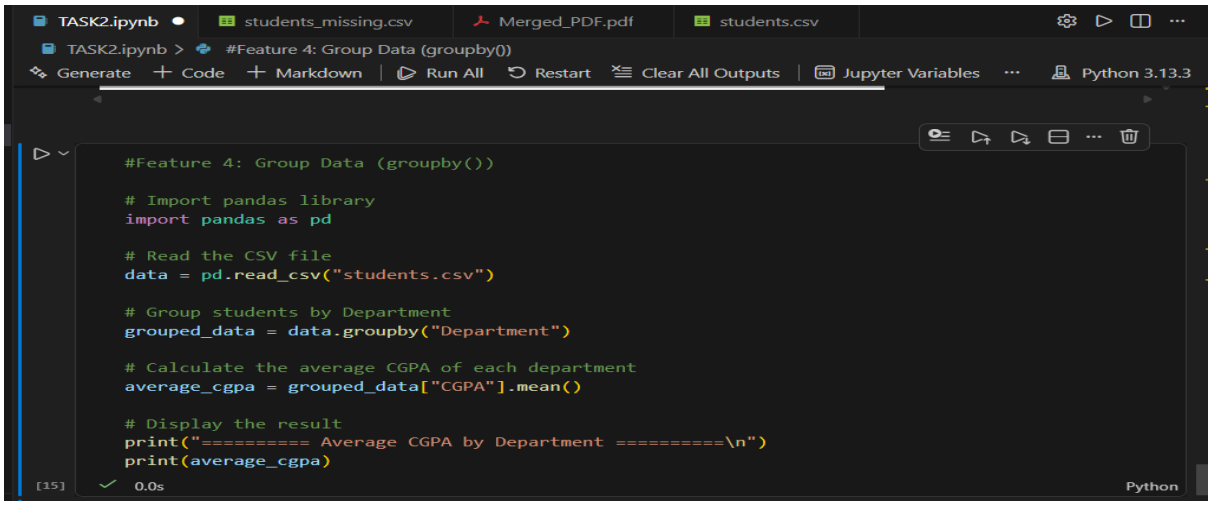
Feature 3: Create graphs and charts (plot())



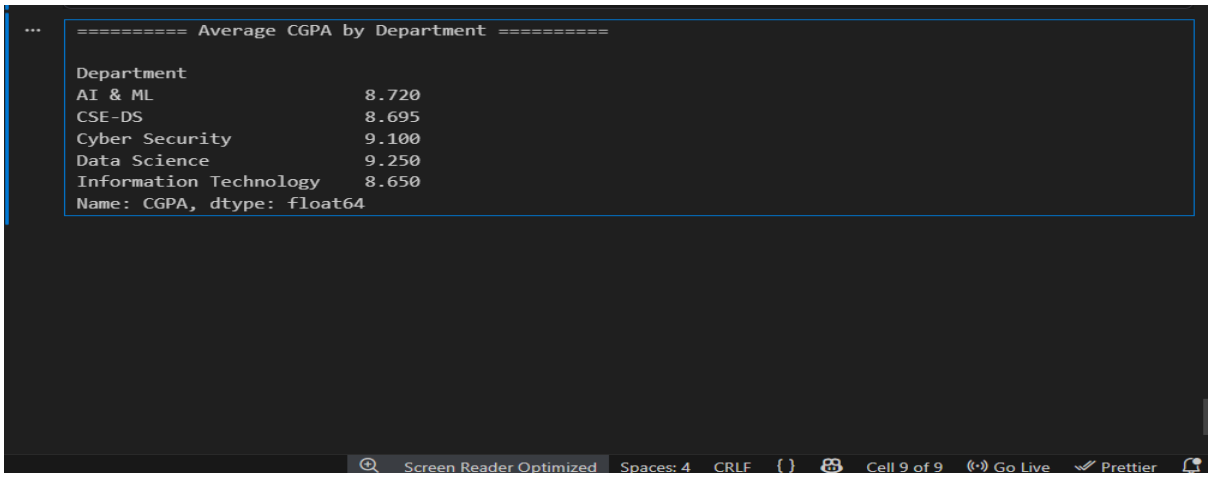
OUTPUT :



Feature 4 : Grouping data(groupby())



OUTPUT :

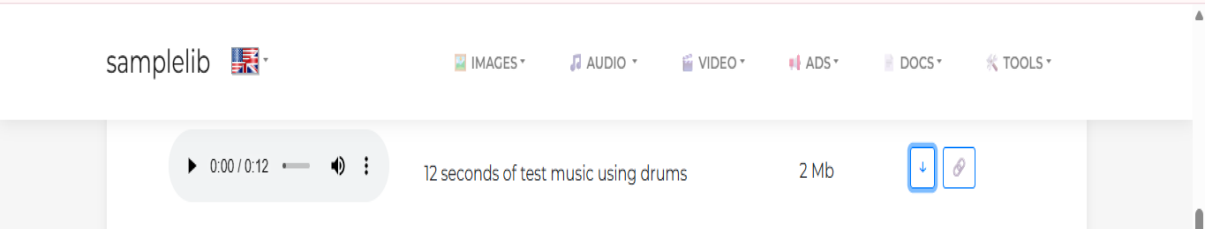


TASK- 3 :

To load an audio file and then convert it into analog file and apply a low-pass filter, and analyze its amplitude andfrequency using Python.

Steps :

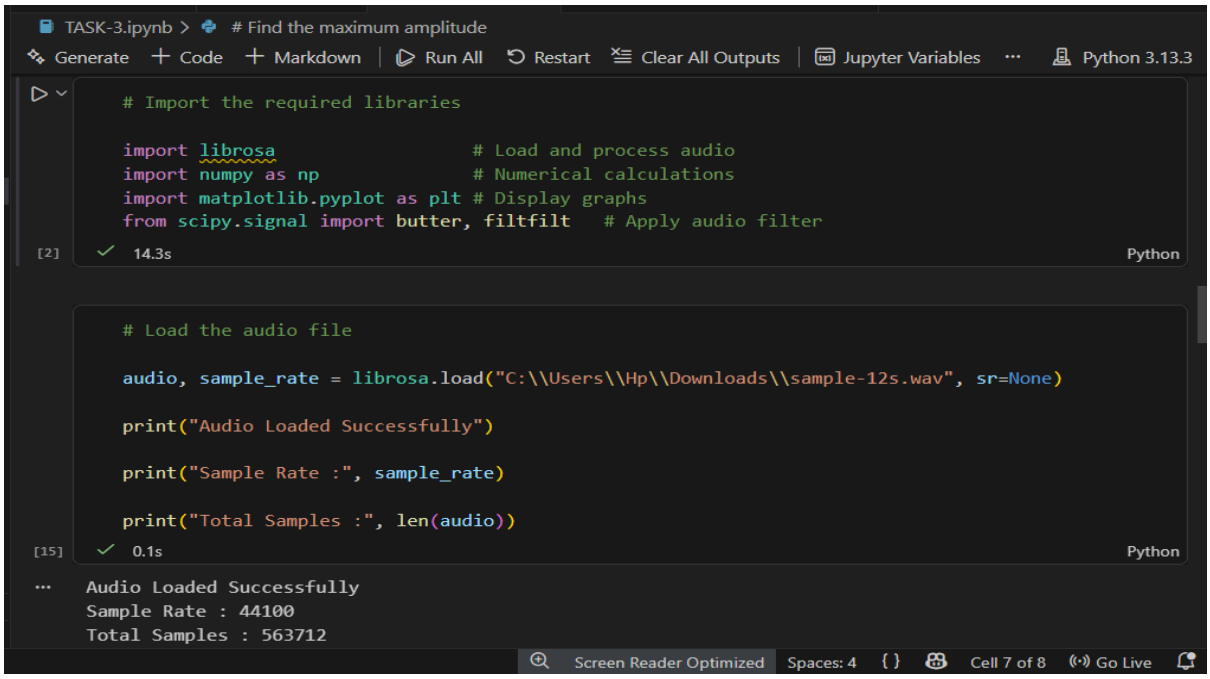
- Download an audio file



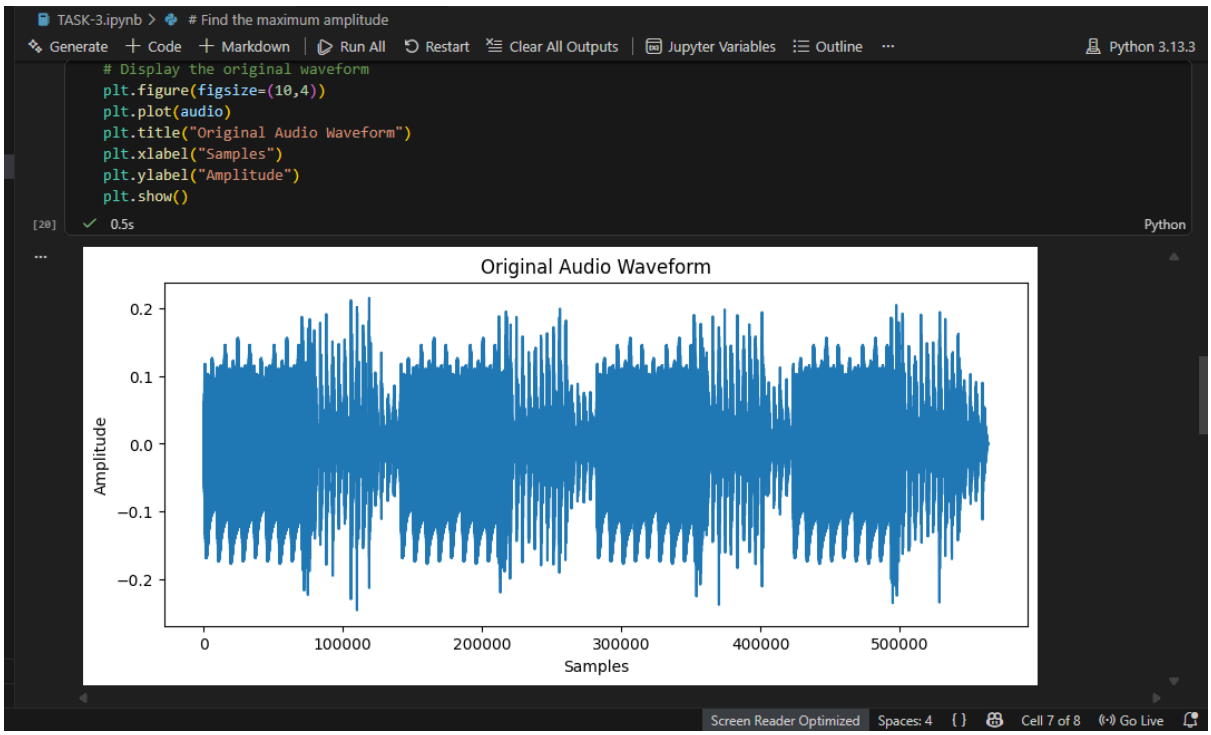
CODE AND OUTPUT :

Step 1: Import Required Libraries

Step 2: Load the Audio File

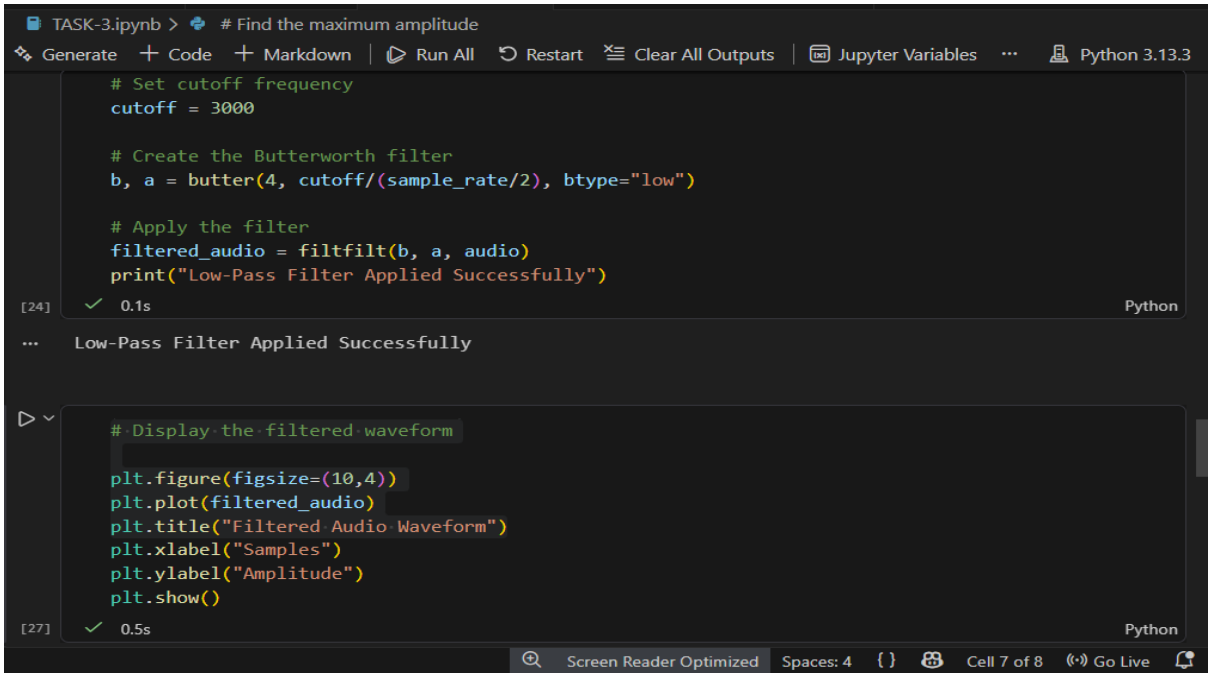


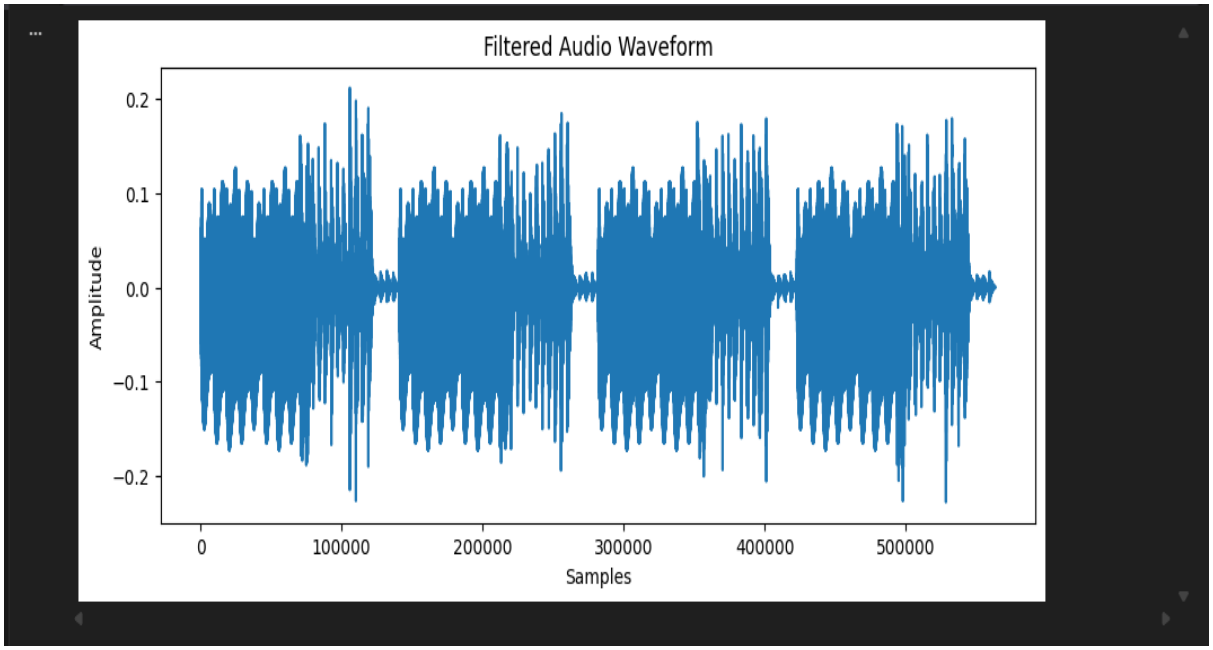
Step 3: Display the Original Audio Waveform



Step 4: Apply a Low-Pass Filter

Step 5: Display the Filtered Audio Waveform

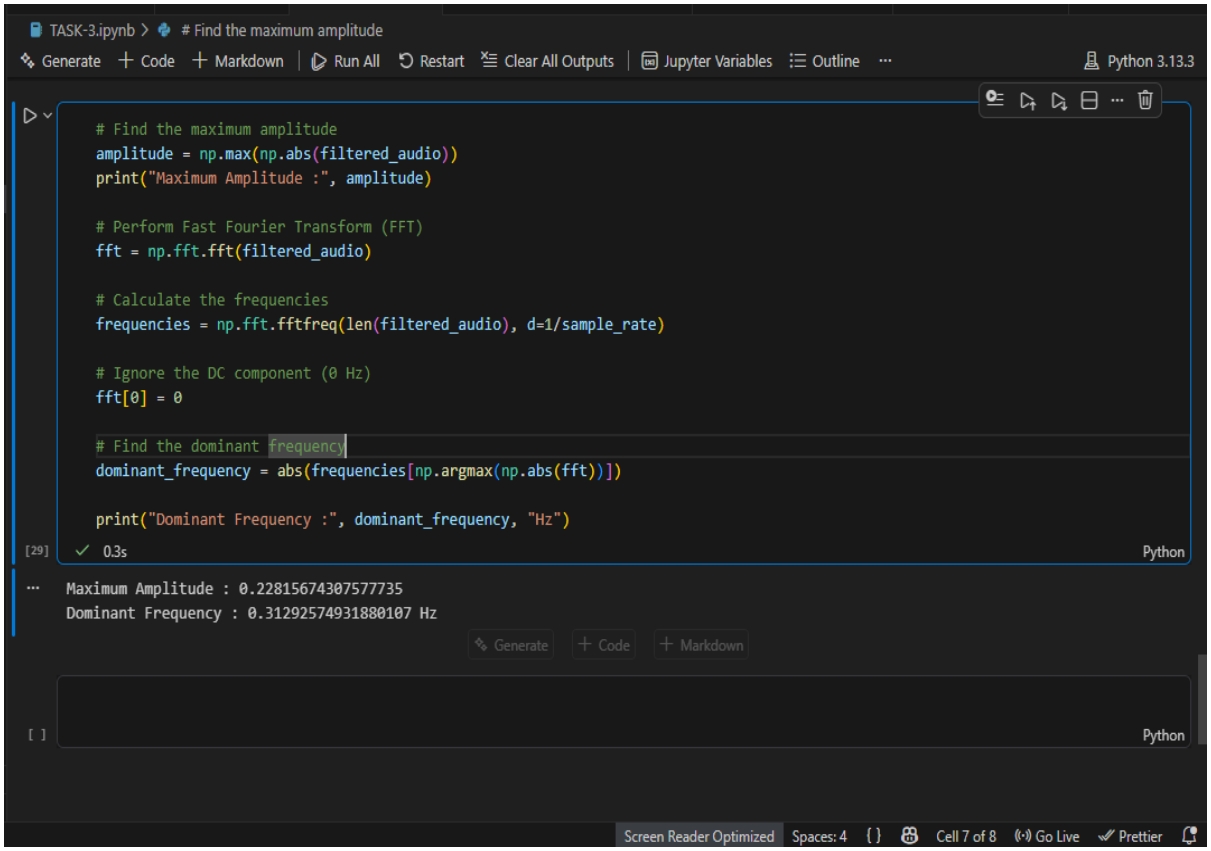




Step 6: Calculate the Maximum Amplitude

Step 7: Perform Frequency Analysis Using FFT

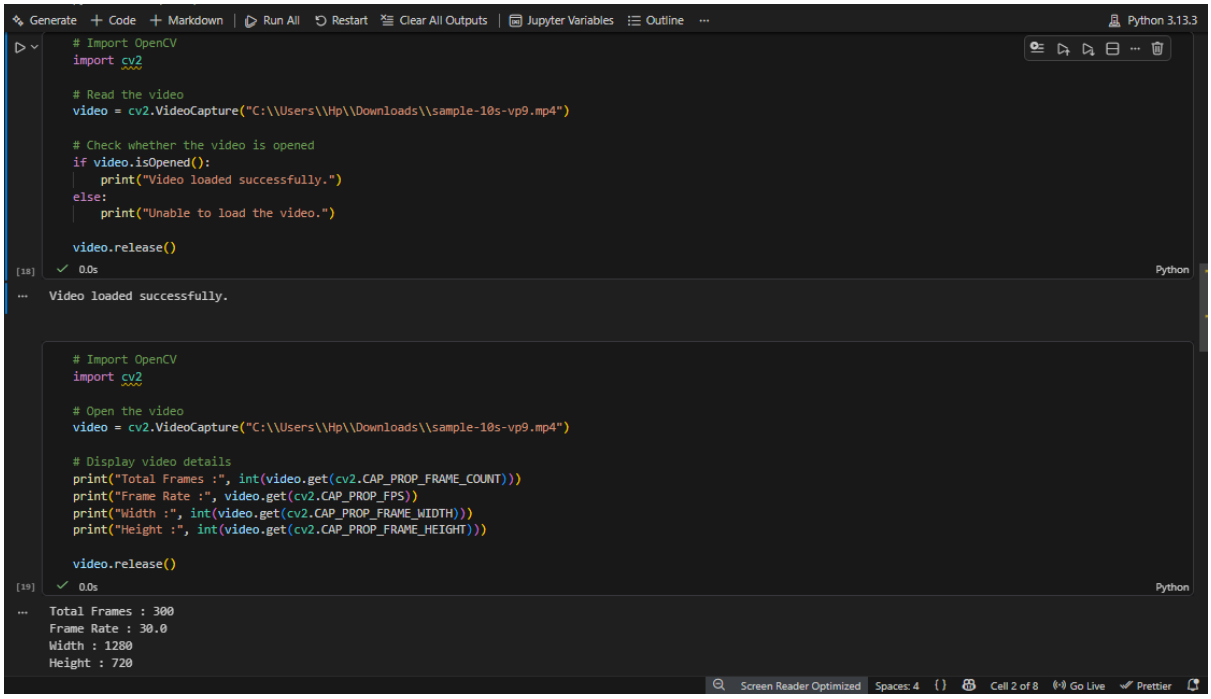
Step 8: Display the Results



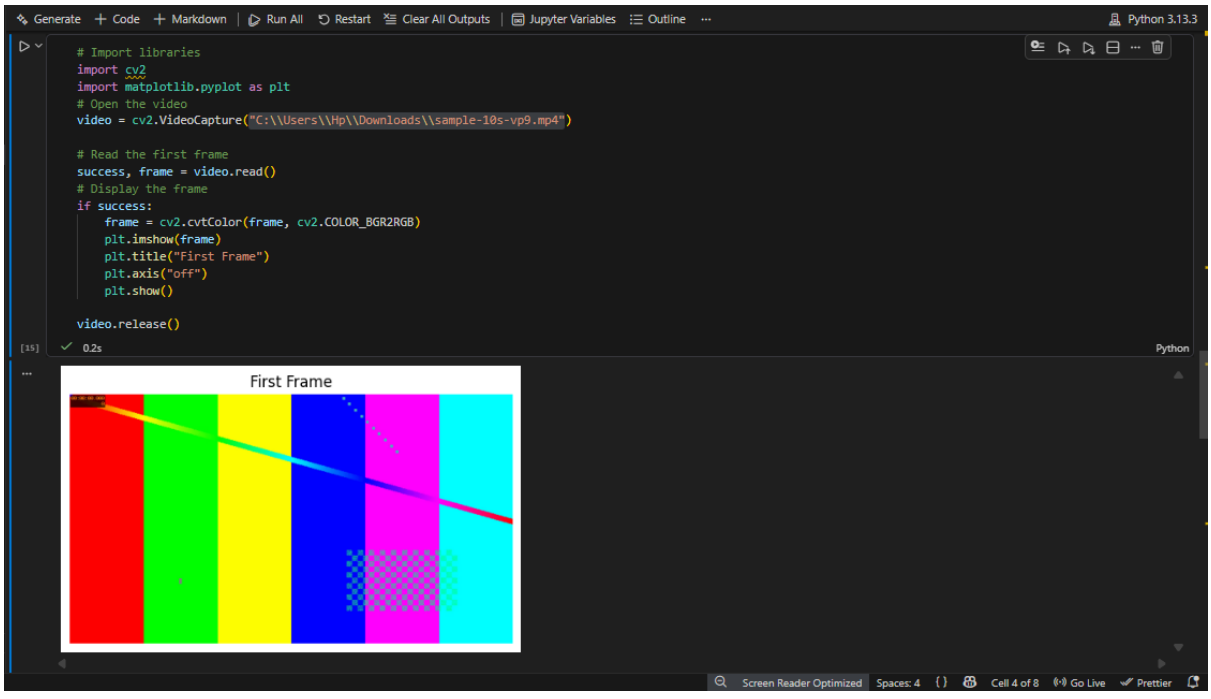
TASK 4: DATA FORMAT OF VIDEO

1. Load the Video File

2.Display the Video Information



3.Extract the First Frame



4.Calculate Maximum Amplitude



```python
# Import libraries
import cv2
import numpy as np

# Open the video
video = cv2.VideoCapture("C:\\Users\\Hp\\Downloads\\sample-10s-vp9.mp4")

# Read first frame
success, frame = video.read()

if success:

    # Convert to Gray
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

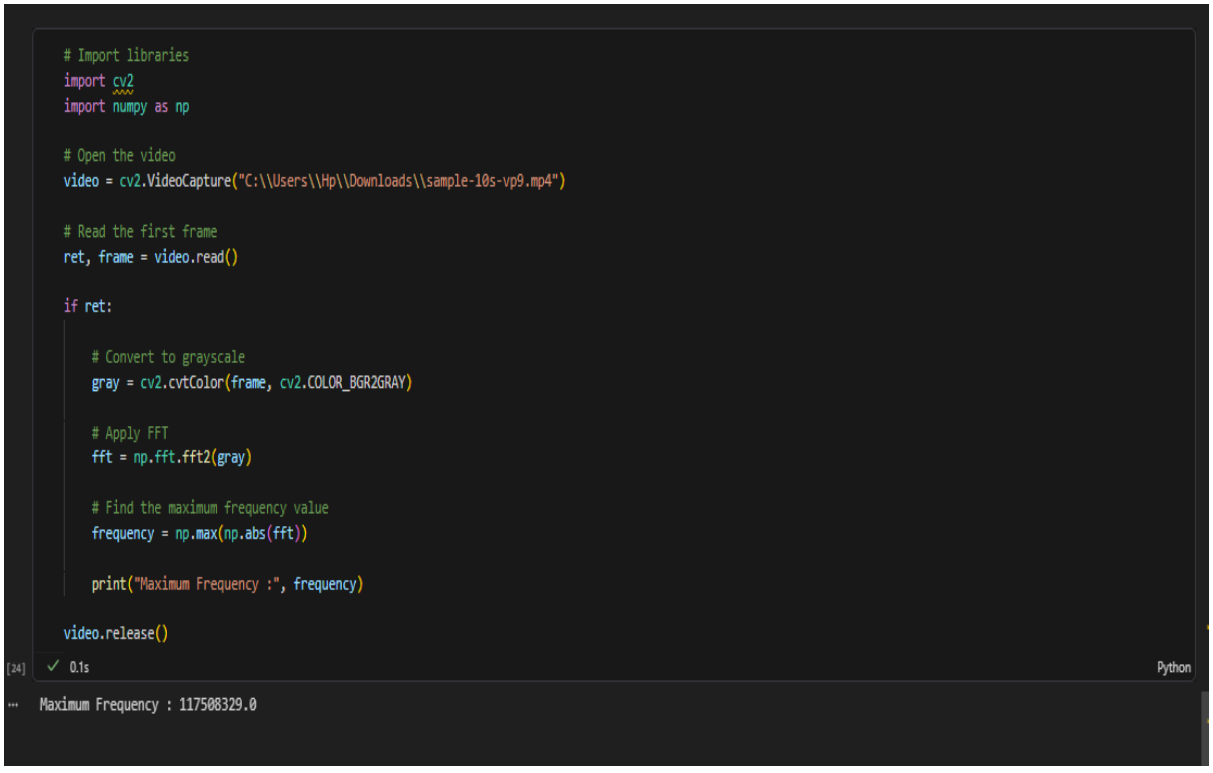
    # Find Maximum Pixel Value
    amplitude = np.max(gray)

    print("Maximum Amplitude :", amplitude)

video.release()
```

```
[23] ✓ 0.0s Python
Maximum Amplitude : 227
```

5. Frequency Analysis



```python
# Import libraries
import cv2
import numpy as np

# Open the video
video = cv2.VideoCapture("C:\\Users\\Hp\\Downloads\\sample-10s-vp9.mp4")

# Read the first frame
ret, frame = video.read()

if ret:

    # Convert to grayscale
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

    # Apply FFT
    fft = np.fft.fft2(gray)

    # Find the maximum frequency value
    frequency = np.max(np.abs(fft))

    print("Maximum Frequency :", frequency)

video.release()
```

```
[24] ✓ 0.1s Python
Maximum Frequency : 117508329.0
```

Task 5: Scraping a Webpage

To access a webpage, verify that it is accessible, extract the webpage content, and calculate the total number of words and characters using Python.

```python
# Import required libraries
import requests
from bs4 import BeautifulSoup
import textwrap

# Enter the webpage URL
url = "https://quotes.toscrape.com"

# Send request
response = requests.get(url)

# Check if webpage is accessible
if response.status_code == 200:

    print("Webpage loaded successfully.\n")

    # Read webpage content
    soup = BeautifulSoup(response.text, "html.parser")

    content = soup.get_text(separator=" ", strip=True)

    print("Webpage Content:\n")

    # Display text neatly (80 characters per line)
    print(textwrap.fill(content, width=80))

    # Count words and characters
    print("\nWord Count :", len(content.split()))
    print("Character Count :", len(content))

else:
    print("Unable to load the webpage.")
```

[26] ✓ 1.0s Python

OUTPUT :

```
Webpage loaded successfully.

Webpage Content:

Quotes to Scrape Quotes to Scrape Login “The world as we have created it is a
process of our thinking. It cannot be changed without changing our thinking.” by
Albert Einstein (about) Tags: change deep-thoughts thinking world “It is our
choices, Harry, that show what we truly are, far more than our abilities.” by
J.K. Rowling (about) Tags: abilities choices “There are only two ways to live
your life. One is as though nothing is a miracle. The other is as though
everything is a miracle.” by Albert Einstein (about) Tags: inspirational life
live miracle miracles “The person, be it gentleman or lady, who has not pleasure
in a good novel, must be intolerably stupid.” by Jane Austen (about) Tags:
aliteracy books classic humor “Imperfection is beauty, madness is genius and
it's better to be absolutely ridiculous than absolutely boring.” by Marilyn
Monroe (about) Tags: be-yourself inspirational “Try not to become a man of
success. Rather become a man of value.” by Albert Einstein (about) Tags:
adulthood success value “It is better to be hated for what you are than to be
loved for what you are not.” by André Gide (about) Tags: life love “I have not
failed. I've just found 10,000 ways that won't work.” by Thomas A. Edison
(about) Tags: edison failure inspirational paraphrased “A woman is like a tea
bag; you never know how strong it is until it's in hot water.” by Eleanor
Roosevelt (about) Tags: misattributed-eleanor-roosevelt “A day without sunshine
is like, you know, night.” by Steve Martin (about) Tags: humor obvious simile
Next → Top Ten tags love inspirational life humor books reading friendship
friends truth simile Quotes by: GoodReads.com Made with ❤ by Zyte

Word Count : 282
Character Count : 1681
```